Collatz Conjecture: Number-and-Fold Closure at Templex One

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Abstract

Take an arbitrary positive odd number. Its Collatz reconstruction determines a unique arithmetic triple in the defect-one fiber: Templex One. The enriched packet additionally depends on the inherited address, provenance and identity conventions. The construction retains the source through three tensor presentations: Atomic retains one addressed member, Micro retains the addressed family, and Macro constitutes that family as one structured whole. Squadron presents these three together. Folding returns one with the fiber reference intact. Under the inherited addressing, projection, constitution and certified-return premises, this composition closes internal Number-and-Fold continuity: Emergent One presents Templex One with fold value one. We derive the exact algebraic relation, carry its members through the scale construction, and perform the return. Five packet-level Snowman declarations have saved Lean and independent-kernel verification; a separate bounded replay checks 100,000 odd sources and four fixtures. Universal finite arrival under ordinary Collatz iteration remains unproved. The distinction is between what the fold preserves and what an arithmetic orbit reaches.

Keywords: Collatz; Snowman; TRYX; ENAID; exact fiber; retained source; typed return; formal verification.

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1. What returns to one?

Start with the distinction that governs the whole construction. A number can reach 1 along an arithmetic path. A structured object can also fold to a value of 1 while retaining the number, its relation and its history. The same numeral appears at the end. We must establish what arrived there.

The ordinary Collatz map on positive integers is

$$C(n) = 3n + 1, \text{ if } n \text{ is odd;} \qquad \text{E01}$$

$$C(n) = n/2, \text{ if } n \text{ is even.}$$

The conventional conjecture asks whether every positive starting value reaches 1 after finitely many applications of C. A commonly used shortcut replaces the odd operation by (3n+1)/2. These conventions have different step counts. We use C for the ordinary map and T_o for the fully accelerated odd map defined below. [1, §1; 2, §1.2]

The literature distinguishes orbit encoding, descent below a starting value, almost-everywhere bounds on orbit minima, and universal arrival at 1. These are different mathematical questions. The comparison below places the present retained-source construction within that literature [1, 11].

The construction uses the author's tensor vocabulary. Atomic, Micro and Macro name presentations of the retained structure at three scales; Squadron presents the three together. At Atomic we retain one source and its reconstruction record. At Micro we retain the family of those records, indexed by source address. At Macro we give the family an explicit constitution specifying membership, identity, boundary and what must survive an operation. Squadron keeps the member, family and constituted whole in correspondence. Section 2 defines these objects before the later sections operate on them. The thundersnow example in Section 2.2 follows one particle through its population and storm, making the nested presentations concrete. [I1, §§6–9; I4, R6–R9]

For Number-and-Fold closure, begin with the complete number-address field. Retain the number through each operation and ask whether the composed return preserves its relation to Templex One. With the inherited premises stated below, the internal closure is

$$\forall\, n > 0\colon C_4(A(n)) \equiv_T \text{TemplexOne}, \qquad \text{E02}$$

$$\text{Return}_{AMM}(C_4(A(n))) \equiv_T \text{TemplexOne};\ \text{FoldValue} = 1.$$

Here A(n) is the canonical complete number-address and C₄ is the composition of Algebraic, Micro, Macro and Squadron closure. It is not the fourth iterate of C. The

statement uses the source's inherited address authority, internal all-number projection, lawful constitution and certified return. Those inputs are part of the result. [I3; I4, R10 and §7; I5]

We will follow the object. Distinguish the exact relation, retain an arbitrary member, and trace that member through the complete family and its constitution. Rotate the presentations together. Then return with the witness attached. If a step loses the source, repair that step before calling the return complete.

This construction follows the September 1 benchmark selected by the September 5 historical lock and its canonical August 24 theorem and internal statement. That choice fixes the definitions used throughout. Later score models and verification records support the exposition at their stated scopes. [I1–I5]

1.1. Relation to the Collatz literature

Stopping time records the first descent below the starting value; total stopping time records the first arrival at 1. Terras [8] and independently Everett [9] established that almost every positive integer descends below its starting value. This historical attribution follows Lagarias's survey [11, §2, Theorem A]. The exceptional set prevents this density-one statement from being a proof for every starting value. Crandall [10] developed early counting results for integers with finite total stopping time, as reviewed in [11, §2, Theorem G]. The later reachability bound [5] concerns this latter question.

Korec [13] strengthened the almost-everywhere control of orbit minima. As stated in Tao's review [2, §1.1], for each $\theta > \log(3)/\log(4)$, the minimum is at most N^{θ} for almost all N in natural density. Tao's theorem allows any bound f(N) tending to infinity, with the conclusion expressed in logarithmic density [2]. Neither result says that every orbit reaches 1. Section 9 compares these quantifiers with all-source packet coverage.

The closest structural comparisons are parity encoding and 2-adic conjugacy [3, 11], inverse trees [4], and predecessor-set analysis [12]. Those works concern iterated dynamics as well as representations. Our exact fiber identifies the one-step odd relation; the retained address and the subsequent constitution require their own stated interfaces. Section 4 compares those objects. Stochastic models provide another perspective: Kontorovich and Lagarias [15] distinguish rigorous results about the models from heuristic predictions about the integer orbits. A regular pattern in a finite plot is therefore not, by itself, an asymptotic theorem for Collatz iteration.

The computational comparison also has two parts: convergence verification over a bounded population [6, 17] and automated reasoning about termination [16]. Section 8 states which predicates the present replay and formal declarations actually check. These comparisons locate the paper's contribution in source retention and its conditional composition; they do not supply the inherited address, projection or certified-return premises.

2. Domains, operators and tensor definitions

Let $O^+=\{1,3,5,\ldots\}$. Each positive integer has a dyadic decomposition $n=2^a H$ with $a\geq 0$ and $H\in O^+$. For an odd source H define

$$p_H = \nu_2(3H+1),\ U_H = T_o(H) = (3H+1)/2^{p_H}, \qquad \text{E03}$$

$$q_H = (H,\ p_H,\ U_H).$$

Remove every factor of two from $3H+1$. What remains is the next positive odd value U_H, and p_H records how many divisions were required. Thus $p_H\geq 1$. This is Tao's Syracuse map; one application accounts for $1+p_H$ ordinary C steps. We keep that count because a change of convention should not quietly shorten the journey. [2, §1.2, (1.1)]

The admissible domain, defect and relation are

$$\Omega = \{(H,p,U):\ H,U\in O^+,\ p\in\mathbb{N},\ p\geq 1\}, \qquad \text{E04}$$

$$D_C(H,p,U) = 2^p U - 3H \in \mathbb{Z}.$$

$$\Gamma_C = \{(H,p_H,U_H):\ H\in O^+\}, \qquad \text{E05}$$

$$\text{TemplexOne} := D_C^{-1}(\{1\}).$$

A fiber is the collection of inputs giving the same output under a specified map. Here $D_C^{-1}(\{1\})$ means every admissible triple whose defect D_C equals 1; this is the fiber named Templex One. The inverse-image notation selects a set of inputs, rather than an inverse arithmetic step. Take this inverse image on Ω. Oddness does real work here: it certifies that all factors of two have been removed. Without it, defect one could describe an unfinished reduction rather than the exact accelerated transition. [I1, §§3–5]

An address α_H=Addr(H) identifies the retained source within the family. The inherited addressing premise requires distinct sources to remain distinguishable by address; this paper does not derive the full address constructor. Provenance Π_H records the source history carried with the object. A reconstruction certificate is the exact equality linking H, p_H and U_H. With these roles distinguished, retain

$$Q_H = \langle \alpha_H,\ q_H,\ 1,\ \text{TemplexOne},\ 2^{p_H}U_H = 3H+1,\ \Pi_H\rangle. \qquad \text{E06}$$

Read the packet from left to right: source address, Collatz point, fold value, fiber, reconstruction certificate and provenance. The address answers which source we retained. The certificate answers whether its reconstruction holds. Requiring a target address α_1 here would introduce a different obligation before deriving it; the canonical successor removes that requirement. [I1, §§6,13; I2]

Arithmetic reconstruction uniquely determines q_H for fixed H. Determining the complete Q_H also requires fixed address and provenance constructors and a convention for identifying certificate records. Those choices are inherited; the dyadic argument does not derive them.

2.1. Atomic, Micro, Macro and Squadron tensors

Here “tensor” refers to the author's addressed, structured presentations used in this construction. The definitions below state the data retained and how the presentations correspond. The operative distinction is between a member, its addressed family, the constitution of that family, and their simultaneous presentation. [I1, §§6–9; I4, R6–R9]

Atomic tensor. Q_H is one complete retained member at the address α_H. Its arithmetic component $q_H=(H,p_H,U_H)$ gives the source, the number of factors of two removed, and the next odd successor. Its other fields retain the fold value 1, the fiber to which the member belongs, the exact reconstruction certificate, and provenance. “Atomic” identifies the member presentation; Q_H is structured data, not just the number H or its fold value.

Micro tensor. $M_{micro}=\{\alpha_H \mapsto Q_H : H \in O^+\}$ is the addressed family of Atomic tensors. The arrow $\alpha_H \mapsto Q_H$ associates an address with its retained member. Thus Micro adds the relation between each member and its place in the complete family. The family is defined by its membership rule over all positive odd H; a finite list or plot illustrates it without enumerating it. Address coverage and distinction remain requirements of the inherited address constructor.

Macro tensor. Mountain is the same family constituted as one structured whole. Its constitution packet K_M specifies the membership domain and predicate, the address and member constructors, the boundary of the field, the common fiber, the identity policy and the provenance to preserve. “Constitution” means these explicit rules for forming and carrying the whole. Micro tells us which addressed members the family contains; Macro states how the family is admitted and retained as one object. Section 4 supplies the constitution and the typed fold/unfold operation. Its all-number projection is an inherited premise, separate from the odd-source arithmetic proved in Section 3. Section 4 states the required recovery of the original positive-number address through the odd-core interface; that bridge is not supplied by the odd-source family alone.

Squadron tensor. $Squadron_AMM=\langle Q_H, M_{micro}, Mountain\rangle$ presents Atomic, Micro and Macro simultaneously; AMM abbreviates their three names. The selected Atomic

member must be the member at its address in Micro, and Macro must carry that same family under K_M. These correspondence requirements give meaning to agreement across scales. Squadron adds their joint presentation; VPA remains the execution sequence within the construction. [I1, §§6–9; I4, R6–R9]

One source across the four presentations. For H=149, 3H+1=448=2^6·7. Atomic retains Q_149, including (149,6,7), its address and the certificate 2^6·7−3·149=1. Micro retains the association α_149↦Q_149 alongside the other admitted members. Macro carries that family with its constitution and common defect-one fiber. Squadron presents Q_149, its family and the constituted whole together. The arithmetic is explicit; the address and whole-family operations retain the premises just identified. This is the same source viewed through the construction, not four unrelated inputs. Section 7 follows its return in detail.

Term	Object used here
Atomic tensor	One complete member Q_H: address, arithmetic triple, fold value, fiber, certificate and provenance
Micro tensor	The complete addressed family $M_{micro}=\{\alpha_H \mapsto Q_H : H \in O^+\}$
Macro tensor	Mountain, the family constituted through its explicit packet K_M
Squadron tensor	The simultaneous tuple of those three presentations
Templex One	The complete fiber $D_C^{-1}(\{1\})$
Emergent One	A returned presentation retaining Q_H and its certified AMM witness
Scalar 1	The fold value, not a replacement for either structured object

VPA is the execution sequence V_n→P_n→A_n→V_(n+1). The source extends this constitution through its addressed population; it does not add a fourth executor. Squadron names all three scale presentations together. We use its agreement and preservation properties after defining the tuple, rather than replacing the tuple by the word “audit.” [I1, §§2,9]

2.2. Thundersnow: one member, nested presentations

Imagine a thunderstorm with snow, and choose an example that also contains hail. Follow one identifiable hailstone through the storm. Atomic, Micro and Macro let us describe the same member within increasingly inclusive structures. Squadron holds those descriptions together. This teaching example gives the nesting a familiar setting before we return to the exact Collatz objects.

Atomic: retain the particle. Select the hailstone and give its record an address. Record its state and formation history. As the particle changes, keep the record connected to that same member. The Collatz counterpart is Q_H: a particular source H with its address, arithmetic triple, reconstruction certificate, fiber and provenance. Atomic means a complete member presentation here; the selected hailstone is itself structured.

Micro: retain the population and its relations. Place the selected hailstone among the snow and hail particles with which it shares the chosen storm region. Retain which member is which, including the selected particle and its recorded interactions. The Collatz counterpart is M_micro, the addressed family containing Q_H. The physical population makes membership and relations visible; the formal family is specified by its rule over every positive odd H, rather than by counting particles.

Macro: constitute the storm as a whole. Specify which population and region belong to the storm being described, how membership is identified, and which records survive a change of presentation. The selected particle remains a member of the population within that whole. The Collatz counterpart is Mountain with its constitution K_M: the rules carrying the family as one structured object, preserving its membership, boundary, identity, common fiber and provenance. The inherited all-number projection remains a separate premise.

Squadron: present all three together. Read the particle record, its population and the constituted storm simultaneously. Check that the particle in the Atomic view is the addressed member in the Micro view, and that Macro contains that same population. Squadron presents those correspondences together. It is not another storm surrounding the first; VPA remains the execution sequence in the formal construction.

Templexed across the presentations. In this example, “templexed” describes keeping the presentations connected through the same retained member and its relations. Before, during and after a recorded change, follow the particle's address and history into the population and the whole. The three descriptions can differ in scale while referring to the same connected structure. This explanatory use of the word adds no new formal operation.

Return now to H=149. The retained particle corresponds to Q_149; its place in the population corresponds to $\alpha_{149} \mapsto Q_{149}$ in Micro; the constituted storm corresponds to Mountain carrying that family under K_M; and the simultaneous view corresponds to Squadron. The actual arithmetic remains $3 \cdot 149 + 1 = 448 = 2^6 \cdot 7$, with defect $448 - 447 = 1$.

These are correspondences between roles, not a claim that hailstones obey the Collatz map. Storm evolution supplies neither the exact defect identity nor the inherited addressing and certified-return premises. The example explains nesting; the mathematical statements and their scope remain those of Sections 2–6. [I1, §§6–9; I4, R6–R9]

Definitions: Atomic · Micro · Macro · Squadron · Templexed

3. Exact algebraic relation and source coverage

Proposition 1 — Unique odd-source arithmetic triple. For each $H \in O^+$, exactly one pair (p,U) with $p \geq 1$ and U positive odd satisfies $2^pU = 3H+1$.

Proof. The integer $3H+1$ is positive and even. Repeated removal of factors of two ends at a positive odd integer. Its maximal exponent p is finite and unique. If $2^pU = 2^qV$ with U,V odd and $p<q$, then U would be even. The case $q<p$ is symmetric. Hence $p=q$ and $U=V$. ■

Proposition 2 — Exact fiber. On Ω,

$$\Gamma_C = D_C^{-1}(\{1\}). \tag{E07}$$

Proof. Membership in Γ_C gives $2^pU = 3H+1$, hence $D_C = 1$. Conversely, $D_C = 1$ gives that same reconstruction equation. Because U is odd, p is precisely $\nu_2(3H+1)$, so the arithmetic triple belongs to Γ_C. Both inclusions hold. ■

The two directions close the algebraic comparison. Every accelerated odd transition has defect one; every admissible defect-one point is that source's unique accelerated transition. Nothing in either argument depends on choosing a small H. [I1, §§4–5; I4, §7 Lemma 1]

The identity $2^p U - 3H = 1$ is the reconstruction equation for the standard accelerated odd map [2, §1.2]. Its defect-one fiber records all admissible one-step transitions. We use Templex One as the name of that structured relation; the name does not make the underlying odd-step equation a new result. The additional content must be located in the retained-address interfaces, constitution and certified return, with their assumptions stated separately.

Corollary — Source projection. The map $\pi_H : \Gamma_C \to O^+$ selecting the first coordinate is bijective, with inverse $H \mapsto q_H$.

Now compare two sources with the same odd successor. Their outputs agree, but their first coordinates still differ. The packet preserves that distinction. Agreement at one field is no permission to erase the others.

The Micro family attaches address and provenance to these points. Under the source's lawful, noncolliding address constructor,

$$\text{Points}(M_{\text{micro}}) = \Gamma_C, \qquad \text{E08}$$

$$\forall\, H \in O^{+}\colon D_C(\text{CollatzPoint}(Q_H)) = 1.$$

Domain, address constructor and member constructor specify the entire family. We can identify an arbitrary member without first enumerating every member. The remaining requirement is precise: Addr must preserve distinct sources. The arithmetic bijection proves source coverage; the address interface must carry that distinction onward. [I1, §6]

Figure 1 maps the finite source/exponent field. For the highlighted U=7 family, H=(7·2^p−1)/3 with even p; successive members satisfy H_(p+2)=4H_p+1. Color gives geometric-mean U within each occupied bin, not source density. The all-source coverage statement above is algebraic; this displayed range is finite.

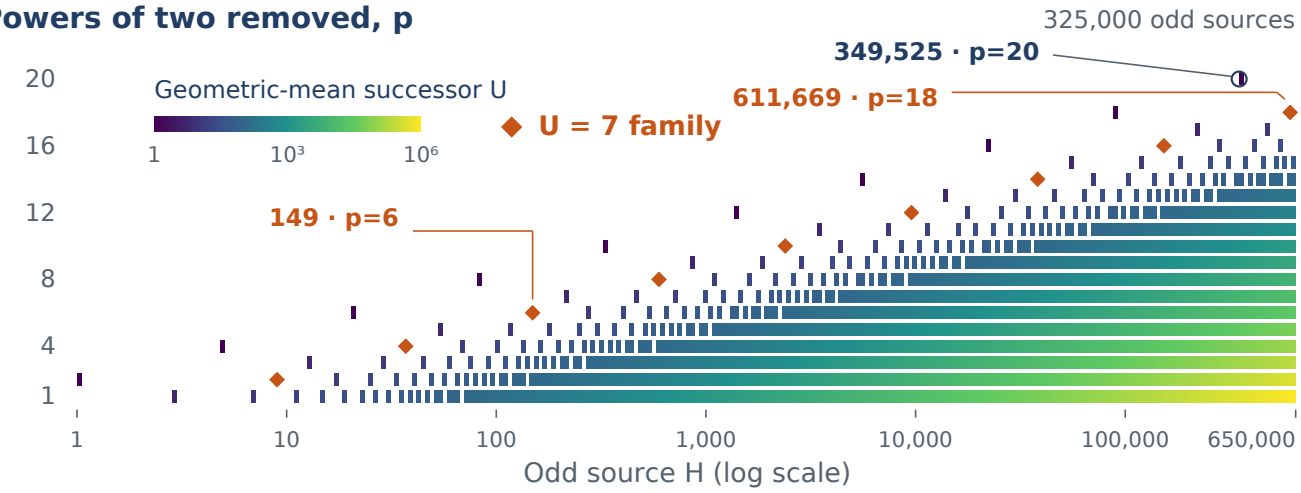


Figure 1. All 325,000 odd sources H ≤ 649999, with p = $\nu_2(3H+1)$. Color summarizes U in 280 logarithmic source bins; orange diamonds mark all nine U = 7 sources. The p = 20 point is included. This finite map is not the complete fiber.

4. Carrying the whole family

The single-source construction now has to carry the complete field. ADDRESS→BIND→CARRY→RETURN states the obligation: identify the number-state, bind its relation, preserve its provenance, and return it. The inherited internal PNP projection supplies the all-number step:

$$A_{\mathbb{N}^+} = \{A(n): n > 0\}, \qquad \text{E09}$$

$$\Pi^{\text{ENAID}}_{\text{P=NP}}(A_{\mathbb{N}^+}) = R_{\mathbb{N}^+}.$$

The output is one relational field specified by constructors, membership and return relation. This step uses the inherited internal PNP lock, simultaneity rule and direct-address constitution. It specifies the complete relation without running a separate solver for each number. It makes no claim that a physical machine evaluates infinitely many inputs for free. [I4, R3, §4.1 and §7 Lemma 3]

Figure 2 makes the need to retain source history concrete. The archived sources 149 and 611669 first share the ordinary state 448 before reaching 7 in 7 and 19 steps. Likewise, 213 and 218453 first share 640 before reaching 5 in 8 and 18 steps. Each common tail is drawn once, with time aligned at its first shared state. Identical continuation does not identify the preceding source history. These finite trajectories motivate provenance retention; they do not implement Fold_M or Unfold_M. Section 7 and Appendix F give the reconstruction records.

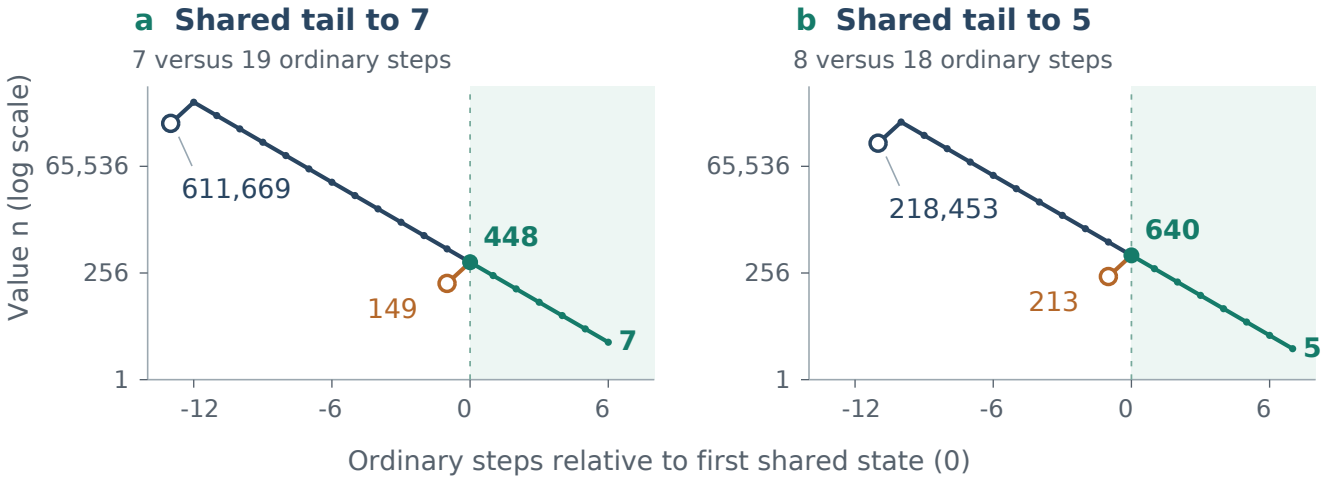


Figure 2. Distinct source histories join at 448 (a) and 640 (b); the identical subsequent states are drawn once in the shaded region. Time is aligned at the first shared state, not at the starts. Open circles mark sources. The current state cannot distinguish these histories. Exact archived fixtures; manuscript §7 and Appendix F.

The exact consumed PNP authority is TRYX.PNP.INTERNAL.SOLUTION.STATEMENT.260824.1437.CANONICAL.LOCK.260824.1438. The current recovered copies of its result/lock lineage include supersession pointers rather than the complete original proof-bearing revision. The present paper preserves the benchmark's declared use of that premise while recording this independent-verification limit. Appendix B specifies the remaining reconciliation. A Boolean assignment theorem elsewhere in the Lean file is not substituted for this projection.

The all-positive bridge in E09 must map each A(n), n>0, to admitted retained data connected to the odd-source family and its K_M, with a recovery map back to that same A(n). The arithmetic decomposition n=2^a H supplies an odd core, but 3, 6 and 12 all have core 3. Recovery therefore requires the original positive-number address and exponent a, or an equivalent source-authorized mechanism retaining the same distinction. This states the required input, output and recovery condition, not a newly established source encoding. The exact bridge and its admission to K_M remain inherited interface obligations; the finite even-source model does not prove this universal map.

A family needs a constitution before we carry it as a Macro object. K_M supplies its membership domain and predicate, Addr, q_H and Q_H constructors, complete-field boundary, common algebraic fiber, identity policy, provenance map, finite specification and algebraic validation. Failure outside lawful admission is explicit. These fields tell us what belongs, what remains distinct, and what must survive the fold. Define

$$\text{Mountain} = \text{Constitute}_{\text{micro} \Rightarrow \text{macro}}(M_{\text{micro}}, K_M). \tag{E10}$$

Follow the same family through the two established equalities: E08 gives Γ_C, and E07 identifies Γ_C with Templex One. Under the constitution rules, Mountain presents that fiber with its retained structure. The point equality establishes which relation we have; lawful constitution establishes how we carry it. Both are needed. [I1, §7; I4, R8]

The locked fold display below is read under an admission and decoder hypothesis: the retained reference must determine the full admitted Mountain, including K_M, addresses and provenance, either through a fixed canonical enrichment or through a uniquely specified referential environment. The extensional set of arithmetic triples alone does not determine those fields. The source invokes a fiber reference [I1, §8; I4, §7 Lemma 2]; matching its exact enrichment and recovery contract remains the typed-decoder obligation in Appendix B. No additional storage mechanism is asserted here. Under that hypothesis:

$$\text{Fold}_M(\text{Mountain}) = \langle 1, D_C^{-1}(\{1\}) \rangle, \tag{E11}$$

$$\text{Unfold}_M(\text{Fold}_M(\text{Mountain})) = \text{Mountain}.$$

Unfold returns the full Mountain only under that recovery hypothesis. A bare numeral 1 is insufficient; a set-only fiber reference recovers the arithmetic relation, not arbitrary enrichment. A single-valued left inverse requires Fold_M to distinguish admitted objects up to their stated identity convention. In the Mountain/Rock terminology, the Rock carries the reference needed for this conditional return. [I1, §8; I4, §7 Lemma 2]

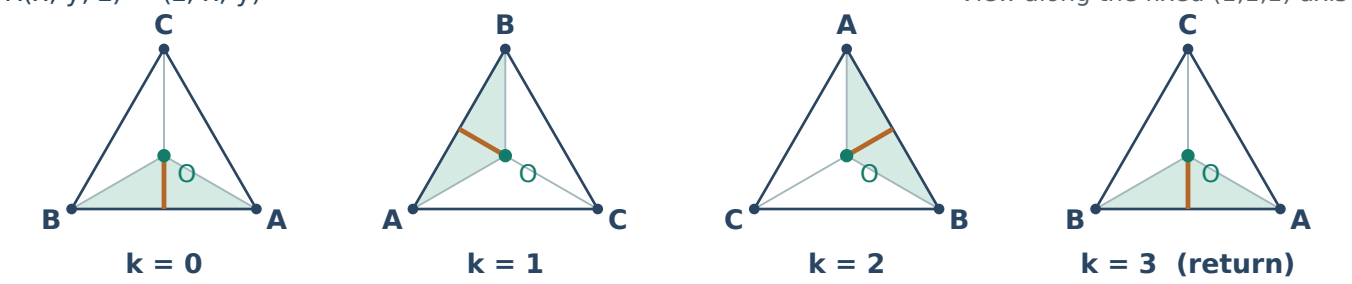


Figure 3. Coordinate model of simultaneous rotation. Face and crease co-rotate about O; three turns restore them. Each view applies at all three scales with the same k. This illustrates the declared model, not full T11 folding. Coordinate model specified in Section 5; E12–E14.

Existing representation theory provides a concrete comparison. Bernstein and Lagarias construct a 2-adic conjugacy between the shortcut Collatz map and the binary shift, with explicit encoding and decoding formulas [3, §1]. Our retained packet and presentation rotation have different declared types; we do not identify them with that conjugacy. Inverse-tree studies instead fix a root and follow repeated preimages [4, §§1–2]. A complete field of one-step packets does not alone prove that every source lies in the inverse tree rooted at 1. Any claimed bridge must supply that additional reachability argument.

The shared tails in Figure 2 are an instance of trajectory coalescence, discussed in Lagarias's account of total stopping times [11, §2]. Wirsching's monograph studies predecessor sets through the Collatz graph and counting methods [12, Chapters II and V]. Our two finite trajectory pairs illustrate loss of source identity after merging; they do not establish a new theorem about the entire predecessor set.

5. Rotate together, preserve the source

Squadron holds the three presentations together:

$$\text{Squadron}_{AMM} = \langle Q_H, M_{micro}, \text{Mountain} \rangle. \qquad \text{E12}$$

At the Atomic tensor, inspect Q_H. At the Micro tensor, locate that same Q_H under α_H in M_micro. At the Macro tensor, inspect K_M and the Mountain it constitutes from that family. Squadron holds all three in view at once, so a change of presentation can be checked against the same retained object. VPA executes the operation. [I1, §9; I4, R9]

Figure 3 illustrates the replay's explicit coordinate model R(x,y,z)=(z,x,y). Let O=(0,0,0), A=(1,0,0), B=(0,1,0), C=(0,0,1). The projection is viewed along the fixed (1,1,1) axis. The source face OAB and the crease from O to the midpoint of AB co-rotate; three turns restore their coordinates. The same k applies at Atomic, Micro and Macro while the model payload stays fixed. This coordinate realization does not establish the full inherited T11 action or decoder premises. [Coordinate model specified here; E12–E14]

Let ρ cycle the roles V→P→A→V, so $\rho^3=\mathrm{id}$ and $\rho^{-1}=\rho^2$. For each scale s, the source defines an anchor-preserving presentation action R_(s,k). Roles, crease and orientation co-rotate. Source number, address, q_H, D_C(q_H)=1, fiber and provenance remain invariant under decoding.

$$R_{\mathrm{Squadron}}(k) = R_{\mathrm{atomic},k} \parallel R_{\mathrm{micro},k} \parallel R_{\mathrm{macro},k},$$
$$R_{\mathrm{Squadron}}(3) = \mathrm{id},$$
$$R_{\mathrm{Squadron}}(-1)R_{\mathrm{Squadron}}(1) = \mathrm{id}. \tag{E13}$$

Three turns here return the presentation. They do not assert that three arithmetic Collatz operations return a source number. Nor does returning a phase label erase an ordinal successor event.

Constitution is co-rotated with its members:

$$R_{\mathrm{macro},k}(\mathrm{Constitute}(M_{\mathrm{micro}},K_M)) = \mathrm{Constitute}(M_{\mathrm{micro}}^{(k)},K_M^{(k)}). \tag{E14}$$

Equation E14 requires the constitution to turn with its members. Leave its presentation coordinates behind and the comparison no longer describes the same operation. Membership, address, point, defect, fiber and provenance must remain fixed under decoding. These preservation and action clauses are inherited premises; Appendix B records the remaining T11/Cross-Scale source matching needed for independent verification. [I1, §§10–11; I4, §7 Lemma 4]

6. Certified Emergent One and closure

A return is complete only when we can say what returned and exhibit the witness binding it to the source. For the benchmark's certified AMM return, retain both:

$$W_H = \langle G_{\to 1}^{\mathrm{AMM}}, Q_H, D_C(q_H)=1, \alpha_H, \mathrm{Provenance}_H \rangle, \qquad \text{E15}$$

$$E_H = \langle \mathrm{TemplexOne}, Q_H, W_H \rangle.$$

The Gattenby return component supplies the three AMM legs. The reconstruction, address and provenance bind those legs to the same source. The benchmark's presentation decoder then yields

$$E_H \equiv_T \mathrm{TemplexOne}, \ \pi_T(E_H) = \mathrm{TemplexOne}, \qquad \text{E16}$$

$$\mathrm{FoldValue}(E_H) = 1.$$

“Emergent 1 = Templex 1” expresses this certified presentation relation. Emergent One carries Q_H and W_H; the decoder presents Templex One, and its fold value is 1. Keeping those fields is what lets us verify the return instead of merely recognizing the numeral. [I4, R5 and §7 Lemma 5; I5]

Internal closure composition. Under the canonical Number-and-Fold definitions and inherited address, projection, constitution and return premises, compose exact algebraic fiber closure, complete Micro coverage, lawful Macro constitution and rotation-stable Squadron presentation. The certified AMM return presents the same fiber. E02 follows only if the all-positive bridge specified in Section 4 retains each original A(n), binds its odd-core presentation to K_M and satisfies the inherited return hypotheses. Section 3 alone proves the odd-source relation, not that complete bridge.

Figure 4 shows the retained-object return described above. Its address, provenance and witness fields also introduce the proposed data-recovery direction: any such application would need to preserve and reconstruct the identified source under a specified decoder. The diagram depicts the mathematical interface; application implementation remains separate.

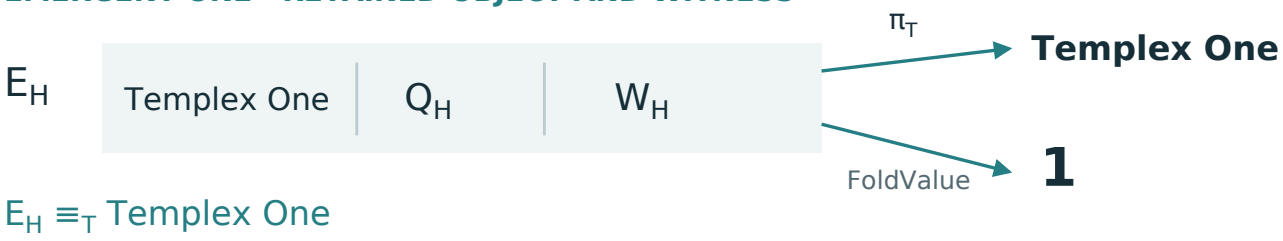


Figure 4. Return with the witness. Emergent One retains Templex One, the source packet Q_H and the AMM witness W_H. The certified presentation decoder yields Templex One; FoldValue yields the scalar 1. Their distinct outputs express the typed identity without discarding the returned object's structure. Full independent verification of the inherited witness specialization remains the obligation stated in §6 and Appendix B.

Application development pending. Figure 4 introduces prospective application directions arising from this construction. Data recovery using the proposed PNP targeting and Collatz addressing framework remains under development. The present paper addresses the mathematical construction and its stated evidence; it does not report an implemented recovery system or establish its performance.

Trace the construction back through its dependencies. Propositions 1–2 establish the relation. The addressed family retains every admitted source. K_M constitutes that family. Co-rotation preserves its specified data. The AMM witness and decoder establish the returned presentation. The internal projection supplies the complete all-number field. Under these premises, internal Number-and-Fold continuity closes. Each operation must carry what the next operation needs. [I1, §12; I3; I4, §7; I5]

The complete witness derivation still needs independent verification. The recovered To1 law leaves an earlier CollatzFold membership clause unproved; the later benchmark invokes a certified specialization. Its exact specialization and external lock must be matched. Writing W_H into a packet identifies the required witness; checking that witness remains a mathematical obligation.

7. Worked performance and arithmetic correspondence

The four archived fixtures expose the reconstruction before giving it a name:

H	p	U	3H+1=2^pU	p=6q+r	D_C
149	6	7	448	6·1+0	1
213	7	5	640	6·1+1	1
218453	17	5	655360	6·2+5	1
611669	18	7	1835008	6·3+0	1

For H=149, begin with 3·149+1=448. Remove the factors of two in order: 448→224→112→56→28→14→7. Six divisions leave the odd carrier 7, so q_{149}=(149,6,7) and $D_C(q_{149})=448-447=1$. The source address remains α_{149}. The number 7 is the next odd arithmetic value; Templex One is the complete defect-one fiber containing this point.

Apply the same calculation to the other three rows. Appendix F lists every halving state, the reconstruction, the shell quotient and remainder, and the ordinary-step count to the carrier. The two sources ending at 5 retain different packets, as do the two ending at 7. The final two source values exceed 199999 and are separately checked fixtures, outside the published 100,000-odd-source sample. These are the four archived reconstruction inputs; the expanded traces are exposition calculated for this edition. [I4, §6; I6, archived_four_reconstructions]

Now carry the 149 packet through the construction. Its address and provenance remain symbolic source fields; no numerical address or fabricated witness is needed to illustrate where they belong.

Presentation / operation	Retained input and intermediate state	Output and justification
Atomic reconstruction	H=149; p=6; U=7; $2^6\cdot7=3\cdot149+1$	$q_{149}\in\Gamma_C$ by Proposition 2; the displayed arithmetic gives defect 1.
Addressed Atomic packet	$Q_{149}=\langle\alpha_{149}$, (149,6,7), 1, TemplexOne, $2^6\cdot7=3\cdot149+1$, $\Pi_{149}\rangle$	The direct source address and provenance remain attached, under the lawful address constructor of §2.

Presentation / operation	Retained input and intermediate state	Output and justification
Micro membership	$\alpha_{149} \mapsto Q_{149}$ in M_micro	This member is included by the family constructor; the other members are retained by the same domain specification.
Macro constitution	M_micro and its full K_M	Mountain= Constitute(M_micro,K_M), under the admission premises of §4. A single fixture does not validate the whole constitution.
Fold and Unfold	Mountain → $\langle 1, D_C^{-1}(\{1\})\rangle$ → Mountain	Under E11's admission/decoder hypothesis, the reference determines the full constituted object. The arithmetic fiber alone does not recover arbitrary enrichment.
Squadron presentation	$\langle Q_{149}$,M_micro,Mountain$\rangle$, with members and K_M co-rotated	E13–E14 preserve the decoded data under the inherited presentation action. This row applies the stated identities; it is not a new executable rotation test.
Certified return	$W_{149}=\langle G_ \to 1\text{^}AMM, Q_{149}$, D_C($q_{149}$)=1, α_{149}, Provenance$_{149}\rangle$; $E_{149}=\langle$TemplexOne, Q_{149}, $W_{149}\rangle$	Under the certified witness and decoder premises, $\pi_T(E_{149})$=TemplexOne and FoldValue(E_{149})=1. The full witness-verification obligation remains as stated in §6.

The arithmetic row is fully calculated. The remaining rows instantiate the framework's retained objects and inherited operations at that same source. They show what each step must carry; they do not replace the required address, constitution or AMM witnesses with labels. The all-number projection E09 remains upstream of the complete-field construction. [I1, §§6–12; I4, R3 and R5–R10]

For an explicit finite arithmetic example, start at $n=12=2^2\cdot3$. The supplied odd chain is $3\to5\to1$: $3\cdot3+1=2\cdot5$ and $3\cdot5+1=2^4\cdot1$. Thus

$$k = 2 + (1+1) + (1+4) = 9, \qquad \text{E17}$$

$$12 \to 6 \to 3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1.$$

This is a newly checked exposition example, not a newly recovered historical performance receipt. In general, a supplied finite odd chain $H_0,\ldots,H_r=1$ yields $k=a+\Sigma_{(i<r)}(1+p_{H_i})$ ordinary steps from $n=2^aH_0$. The formula decodes an existing finite witness. It does not establish the existence of such a witness for every n. The two initial halvings reach 3; the first odd block uses $3\to10\to5$, and the second uses $5\to16\to8\to4\to2\to1$. Appendix F records the nine ordinary transitions individually. We stop at the first displayed arrival at 1; C itself is not being redefined to make 1 an absorbing state.

Run the same comparison at each scale. At Atomic, show the addressed source and its reconstruction. At Micro, show where it belongs in the complete family. At Macro, show the constitution and retained fiber. At Squadron, rotate the presentations together and inspect the certified return. The formulas must do the work on the page. Appendix A connects these operations to the original R-stages; later M-measure labels provide a separate exposition crosswalk.

Figure 5 expands the supplied path from 12 to 1 into nine ordinary Collatz steps: two initial halvings, two steps from 3 to 5, and five steps from 5 to 1.

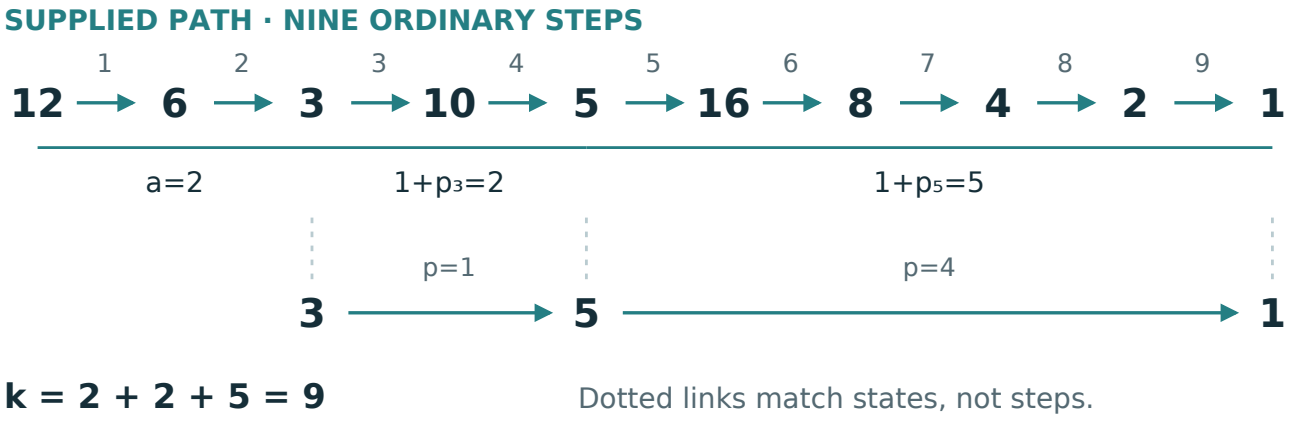


Figure 5. Count the steps in a supplied path. The initial dyadic reduction of 12 uses two ordinary steps. The odd transitions $3\to5$ and $5\to1$ expand into two and five ordinary steps, giving nine in total. This finite example illustrates the decoder for an existing path; it does not establish a finite path for every starting value.

Applegate and Lagarias [18] study total stopping times for the shortcut map and prove the existence of infinitely many inputs with long but finite trajectories. Their asymptotic

result concerns a different scale from the four archived examples here. Our counts to the odd carrier are finite ordinary-step counts, not total stopping times to 1 or estimates of their distribution.

8. What survives checking?

The formal checks begin with a smaller object: Packet retains source, power, odd carrier, positivity and oddness proofs, and reconstruction. Its foldValue uses natural-number subtraction; the reconstruction proves the value is one on admitted packets. This tests a definite part of the construction. The complete Q_H, K_M and W_H objects remain outside that module. [I6]

Snowman declaration	Checked statement
positive_dyadic_ factorization	Every positive natural m admits 2^p·u=m with u positive odd
packet_fold_one	Every admitted Packet has foldValue 1
all_odd_addresses_ have_fold_one	Every positive odd h has a Packet retaining source h and fold value 1
different_sources_ keep_distinct_packets	Unequal source coordinates imply unequal Packet records
archived_four_ reconstructions	The conjunction of the four numerical reconstruction equations above

The first and third statements are universal within their stated types. They are not finite tests. Factorization uniqueness and the full fiber biconditional are not separately submitted declarations. The Snowman proofs do not invoke the co-located NS or PNP theorem bodies.

Saved run 34786022732, attempt 2, records successful compilation and independent checking of these five targets within an eleven-target project. The independent log records acceptance by nanoda and Lean's default kernel. The first three Snowman results report propext and Quot.sound; the other two report propext. Classical.choice is permitted by configuration but is not reported for those five results. The challenge's eleven intentional placeholders are distinct from the solution's proof bodies. Challenge and solution share their specification, so kernel independence is not independent authorship of the mathematical formulation. [I6; V1]

The published Python replay checks 100,000 odd sources from 1 to 199999 and the four fixtures. It reports zero recorded failures. Generation and checking share arithmetic, and its AMM certificate uses Boolean combinations of defect, reconstruction and shell predicates. The strings identifying EmergentOne and TemplexOne are assigned by its constructor and then checked. These checks support the serialized representation; they do not independently derive the full symbolic witness or its decoder. [V2]

Beta 3 is separate development evidence: 1,000 positive inputs, 20,000 finite fiber comparisons, 26 checks and 54 control outcomes. Some controls reject malformed records; others compare orientation identities or distinguish lawful regroupings. Its coordinate rotation and paired-view schema are declared models, not a formal proof of every inherited source operation. Appendix D retains those distinctions. [V3]

Computational convergence verification tests a different property from these packet checks. Barina's 2025 article reports convergence verification up to 2^71 [6]. Our reported population measures reconstruction and representation checks under the specified schema; it is not a competing convergence bound. Lean 4 supplies the proof-assistant infrastructure [7], while the exact declarations and retained checking records specify this project's formal coverage.

Barina's earlier algorithmic paper [17] reduces the size of lookup tables used in convergence verification; the later work [6] develops the larger verification campaign. These are algorithmic and computational results about orbit convergence. Yolcu, Aaronson and Heule [16] instead construct a string-rewriting system whose termination is equivalent to the Collatz conjecture, and obtain automated proofs of nontrivial weakenings. They do not prove the full conjecture. For the present formalization, a corresponding termination claim would require an explicit bridge from the checked packet declarations to finite arithmetic arrival.

9. Where the argument stops

Return to the opening question. We have the complete algebraic relation and the internal continuity composition under its inherited premises. A temporal orbit asks for something further: a finite chain ending at 1 for each starting value. The canonical theorem and selected lock retain that universal statement as unproved. Therefore E02 and

$$\forall\, n > 0\ \exists\, k \in \mathbb{N}\colon C^{k}(n) = 1. \qquad \text{E18}$$

state different results. A direct address locates the source; a finite first-HIT certificate exhibits its arrival. Accepting a supplied terminal certificate does not establish that every input has one. This boundary belongs to the original closure statement. [I1, §14; I4, §7 temporal boundary and §8; I5]

The source's open entries include receiver-local address origin, uniform zero-cycle exclusion, separated refill compensation bound, universal first negative crossing and universal terminal certificate/finite termination. Inherited-source reconciliation also remains for the exact PNP revision, To1 specialization, Cross-Scale admission and rotation clauses, and the original supplied-chain adjudication. Missing ancestor bodies, an unimplemented Lean target and an unproved mathematical proposition are different limitations. Appendix B keeps them separate.

For comparison with conventional literature, Tao's theorem gives, for every function f tending to infinity, a minimum orbit value below f(N) for almost all N in logarithmic density. Its quantifiers are neither a finite sample nor universal arrival at 1. It provides context for the temporal problem, not a premise of the internal typed closure. [2, Definition 1.2 and Theorem 1.3]

Krasikov and Lagarias prove, by a computer-assisted argument, that at least x^0.84 integers below sufficiently large x contain 1 in their forward orbit [5]. This counts actual arrival at 1. That reachability statement, Tao's almost-bounded minimum theorem, and the present one-step fiber representation concern different properties.

Cycle exclusion is a separate part of that boundary. Simons and de Weger [14, Theorem 3(b)] rule out nontrivial m-cycles for $1 \le m \le 68$ in their 2005 analysis. Here m counts pairs of increasing and decreasing blocks in their shortcut-map convention, not individual ordinary steps. This is a historical bound rather than a claim about the latest record. The three-step return of a presentation rotation in Section 5 is not a cycle of the arithmetic Collatz map, so it does not discharge this dynamical obligation.

10. Conclusion

Begin with an arbitrary odd source and reconstruct it exactly. Its arithmetic component belongs to the complete defect-one fiber. Retain its address and certificate in the Micro family, carry that family through Macro constitution, and present all three scales together in Squadron. The certified AMM return then brings the retained object back to Templex One under the inherited premises.

The internal conclusion is EmergentOne≡_T TemplexOne with fold value 1. Source, relation and witness remain available for inspection. The Lean packet results and bounded performances verify their stated parts; the full inherited interfaces and universal finite-orbit question retain their open obligations. Closure has a scope. Keeping it visible is part of completing the return.

Authorship, assistance and provenance

Virgil Lee Gattenby is the identified human author and project decision authority. Original source headers use “Virgil / TRYX-ENAID program.” AI assistance includes computational packaging, Lean translation, source recovery, evidence mapping and preparation of this manuscript. Publication 9.24 corrects the edition labels at the author's direction on November 7, 2026. The prior edition is retained.

The August 24 construction and internal statement precede the September 1 benchmark selected by the September 5 lock. The benchmark's original header remains unchanged. Its displayed timestamp is not treated as completion time of all later corrections. The separately retrieved public release v1.0.0 has published_at 2026-09-11T06:12:04Z and a tag target different from the later pinned formal-source commit. This establishes recorded edition events, not first public disclosure or elapsed research time. [I1–I5; P1–P4, P6]

This publication correction makes no new manuscript/code license grant or claim of journal acceptance or peer review. Source-specific attribution and dependency licenses remain recorded in the permissions register. Mathematical scope is unchanged.

Appendix A. Source-to-paper concordance

Original stage	Manuscript location	Later draft crosswalk
R1 — Answer over time	§§1,7,9	M0 introduction; stage distinctions
R2 — VPA rotation	§§2,5	M8, with model scope retained
R3 — PNP spine and projection	§4	Spans the construction; no standalone draft token substitutes
R4 — Algebra first	§§2–3	M1–M3
R5 — Templex/Emergent return	§§2,6	M6/M9 typed return
R6 — Atomic number assembly	§§2–3,7	M4
R7 — Micro closure	§§3,7	M4–M5
R8 — Macro closure	§4	M5–M6
R9 — Squadron	§5	M8
R10 — Terminal typed identity	§§1,6,10	M9
R11 — Finite performance	§§7–9	Evidence scope, distinct from source proof

M7's later nested-handoff tree is supplementary exposition, not an additional original R-stage. Draft equation labels E01–E18 are distinct from the historical Beta measure labels M0–M9 and from bibliography entries [1]–[18] ([1] and [2] were formerly E1/E2). Original source identifiers remain unchanged in the inheritance record.

An internal editorial inventory maps 107 source units to manuscript destinations. That inventory and the retained source files are archival records, not additional deliverables. Appendix H distinguishes publicly retrievable evidence from internal review records. No original formula is edited to fit a page count.

Appendix B. Assumptions and dependency frontier

Interface	Source premise and remaining verification
Address/family	Lawful ENAID address, identity and provenance contracts; direct source address is distinct from finite first-HIT address
Internal projection	Original PNP 1437/1438 result/lock, current-state simultaneity rule and 1409 constitution; original revision and subsequent amendment scope require reconciliation
AMM return	GATTENBY.TO1.LAW.260824.1126 and its cited lock; exact Collatz specialization and all three return legs require matching to the later benchmark witness
Typed decoder	Locked public presentation-equivalence contract cited by the benchmark; matched original body/lock and full-enrichment recovery environment remain a recovery task
Constitution/rotation	Cross-Scale Law 2, T11 and VPA-only execution; bind each admitted field/action to its exact inherited clause
Finite certificate architecture	Original August 30 adjudication and its hypotheses; universal certificate existence remains separate

The benchmark's historical resident-source claims are part of its record. They do not recover unavailable bodies in the present packet. The source's declared internal closure and the present independent-verification frontier are therefore reported separately. [V4]

Appendix C. Formalization and reproduction

Pinned formal source: TRYX-Relay/TRYX-Collatz-Mathic, commit c80c82450cbbeb9148f5a78189b307a3336111fd, verification/tryx-lean. Toolchain: leanprover/lean4:v4.34.0-rc2. Mathlib: 85e3a25e006c35636f0e53b0e9296caca2685bc0. Comparator: 19e111e2141cf333c7daff0f64c5f24acc91dd2e. The manifest records eleven Lake dependencies; the independent log additionally identifies nanoda_lib 4c544ed4099c8227f07d5de77ad1e69fb0740a27 and landrun 811cfff51ceaf3d9843708aa6d22e9b84ccac8b4.

Saved operations include lake build TryxProof, lake env lean TryxProof.lean, and lake exe comparator comparator.json within the recorded checking environment. Build job 112478265699 and independent job 112479278510 both report success. These command

names locate the operations; the retained logs and manifest supply setup details. They are historical receipts, not executions performed while writing this draft. [V1]

The published replay operation is python3 replay.py, recorded with exit code zero. Its receipt binds script and output hashes but does not record a Python version. Beta 3's separate receipt records Python 3.9.6. No interpreter version is inferred across those receipts. [V2–V3]

Proposed formal extensions include unique factorization, the full fiber biconditional, source bijection, addressed family, typed fold/unfold, K_M constitution, rotation/equivariance and the full AMM return/projection composition. These are proposed targets, not additional checked declarations. Their implementation belongs in a successor submission. [V5]

Appendix D. Finite evidence and preserved repair history

Record	Population / status
Original benchmark	Source-reported 100,000 odd-source field and four fixtures
Published replay receipt	100,000 odd sources and four separately checked fixtures; PASS
September 8 historical receipt	Separate 4,096-member ubiquity, 10,000-start trajectory and nested campaigns; separate 100,000-start trajectory replay; reported PASS
Beta 3	1,000 inputs; 20,000 comparisons; 26 checks; 54 control outcomes; finite model PASS

These counts describe distinct campaigns and are not added into one purported coverage population. The published replay has no deliberate corrupted-input suite. Beta controls test packet/domain mutations, address/schema/provenance, handoff, frame/rotation, paired views and orientation. Shared validators limit independence. The lawful-regrouping control checks structural distinction rather than rejecting a valid alternative.

The selected benchmark reports an original D3 failure retained before its Micro repair checkpoint, and an earlier ASCII repair with an n=7 fixture. The original failed packets have not been inspected in this recovery. Their precise causes are not reconstructed from later success. The original and successor records must both be retained if those mechanisms are described further. [I4, §5; V6]

Appendix E. Topology conventions

A later shared-apex record separates conditional geometric completion from the designation of its opposite face as 1_E. For $B_2=[OAB]+[OBC]+[OCA]-[ABC]$, the standard alternating boundary convention gives $\partial[OABC]=-B_2$. The saved Beta successor uses $-[OABC]$ to realize B_2. This orientation qualification applies to that later topology formula; it neither modifies the original Collatz bytes nor promotes the disputed September 8 reinterpretation to governing authority. Certain cross-domain decoder/operator definitions remain unfinished or unresolved. [V7]

Appendix F. Complete arithmetic traces for the worked examples

The four source triples are historical fixtures from I4 §6 and I6. The expanded intermediate traces below are newly calculated exposition. Each sequence begins at 3H+1 and records every division by two until the next odd value. The count to that carrier includes the initial odd operation. None is asserted to be the total stopping time from H to 1.

F.1 — Source 149

Input H=149. Odd operation: 3·149+1=448.

Halving states: 448 → 224 → 112 → 56 → 28 → 14 → 7.

There are 6 divisions, ending at odd U=7; hence $p=\nu_2(3H+1)=6$. Reconstruction: 2^6·7=448. Defect: 448−447=1. Shell accounting: 6=6·1+0, with 0≤0<6. Ordinary steps from H to this carrier: 1+6=7. Retained point: (149,6,7); direct address: α_149.

F.2 — Source 213

Input H=213. Odd operation: 3·213+1=640.

Halving states: 640 → 320 → 160 → 80 → 40 → 20 → 10 → 5.

There are 7 divisions, ending at odd U=5; hence $p=\nu_2(3H+1)=7$. Reconstruction: 2^7·5=640. Defect: 640−639=1. Shell accounting: 7=6·1+1, with 0≤1<6. Ordinary steps from H to this carrier: 1+7=8. Retained point: (213,7,5); direct address: α_213.

F.3 — Source 218453

Input H=218453. Odd operation: 3·218453+1=655360.

Halving states: 655360 → 327680 → 163840 → 81920 → 40960 → 20480 → 10240 → 5120 → 2560 → 1280 → 640 → 320 → 160 → 80 → 40 → 20 → 10 → 5.

There are 17 divisions, ending at odd U=5; hence $p=\nu_2(3H+1)=17$. Reconstruction: 2^17·5=655360. Defect: 655360−655359=1. Shell accounting: 17=6·2+5, with 0≤5<6.

Ordinary steps from H to this carrier: 1+17=18. Retained point: (218453,17,5); direct address: α_218453.

F.4 — Source 611669

Input H=611669. Odd operation: 3·611669+1=1835008.

Halving states: 1835008 → 917504 → 458752 → 229376 → 114688 → 57344 → 28672 → 14336 → 7168 → 3584 → 1792 → 896 → 448 → 224 → 112 → 56 → 28 → 14 → 7.

There are 18 divisions, ending at odd U=7; hence $p=\nu_2(3H+1)=18$. Reconstruction: 2^18·7=1835008. Defect: 1835008−1835007=1. Shell accounting: 18=6·3+0, with 0≤0<6. Ordinary steps from H to this carrier: 1+18=19. Retained point: (611669,18,7); direct address: α_611669.

F.5 — Supplied finite path from 12

Input $n=12=2^2\cdot 3$. Initial dyadic exponent a=2. Odd chain 3→5→1 has $p_3=1$ and $p_5=4$. The sum $a+(1+p_3)+(1+p_5)=2+2+5=9$ agrees with the explicit steps:

Step	Input	Operation	Output	Block
1	12	n/2	6	Initial reduction
2	6	n/2	3	Initial reduction
3	3	3n+1	10	3→5
4	10	n/2	5	3→5
5	5	3n+1	16	5→1
6	16	n/2	8	5→1
7	8	n/2	4	5→1
8	4	n/2	2	5→1
9	2	n/2	1	5→1

This is a finite witness for the starting value 12. The general step decoder applies to a supplied finite odd chain ending at 1; it does not prove that such a chain exists for every positive input. The explanatory figures accompanying §7 visualize these same states and do not add proof premises.

Appendix G. Definitions and navigation

Click a linked term in the paper to open its definition. Each entry links to its main use; Contents provides the section map. The definitions describe the construction and its stated premises. [I1, §§6–9; I4, R2–R9]

Atomic tensor

Q_H is one complete retained member at the address α_H. Its arithmetic component q_H=(H,p_H,U_H) gives the source, the number of factors of two removed, and the next odd successor. Its other fields retain the fold value 1, the fiber to which the member belongs, the exact reconstruction certificate, and provenance. “Atomic” identifies the member presentation; Q_H is structured data, not just the number H or its fold value.

Main use: Section 2 · Thundersnow: Section 2.2 · Contents

Micro tensor

$M_{micro}=\{\alpha_H \mapsto Q_H : H \in O^+\}$ is the addressed family of Atomic tensors. The arrow $\alpha_H \mapsto Q_H$ associates an address with its retained member. Thus Micro adds the relation between each member and its place in the complete family. The family is defined by its membership rule over all positive odd H; a finite list or plot illustrates it without enumerating it. Address coverage and distinction remain requirements of the inherited address constructor.

Main use: Section 2 · Thundersnow: Section 2.2 · Contents

Macro tensor

Mountain is the same family constituted as one structured whole. Its constitution packet K_M specifies the membership domain and predicate, the address and member constructors, the boundary of the field, the common fiber, the identity policy and the provenance to preserve. “Constitution” means these explicit rules for forming and carrying the whole. Micro tells us which addressed members the family contains; Macro states how the family is admitted and retained as one object. Section 4 supplies the constitution and the typed fold/unfold operation. Its all-number projection is an inherited premise, separate from the odd-source arithmetic proved in Section 3. Section 4 states the required recovery of the original positive-number address through the odd-core interface; that bridge is not supplied by the odd-source family alone.

Main use: Section 4 · Thundersnow: Section 2.2 · Contents

Squadron tensor

Squadron_AMM=⟨Q_H,M_micro,Mountain⟩ presents Atomic, Micro and Macro simultaneously; AMM abbreviates their three names. The selected Atomic member must be the member at its address in Micro, and Macro must carry that same family under K_M. These correspondence requirements give meaning to agreement across scales. Squadron adds their joint presentation; VPA remains the execution sequence within the construction. [I1, §§6–9; I4, R6–R9]

Main use: Section 5 · Thundersnow: Section 2.2 · Contents

Templexed: connected presentations

In the thundersnow teaching example, this word describes keeping Atomic, Micro and Macro connected through the same retained member, its relations and its history. Squadron presents those correspondences simultaneously. This is an explanatory usage; the formal construction still depends on its specified packets, constitution and operations.

Worked explanation: Section 2.2 · Contents

Address

A source identifier α_H=Addr(H) used to locate Q_H in the Micro family. Distinct admitted sources must remain distinguishable. The full constructor is an inherited premise.

Main use: Section 2 · Contents

Provenance

The retained source history Π_H carried with a packet through the construction. It is distinct from the numerical fold value.

Main use: Section 2 · Contents

Reconstruction certificate

The equality 2^p_H U_H=3H+1 carried with Q_H. It checks the exact odd-source reconstruction; it does not certify universal finite arrival at 1.

Main use: Section 2 · Contents

Fiber and Templex One

A fiber collects all inputs giving a specified map value. Templex One is $D_C^{-1}(\{1\})$ on Ω: the admissible odd-source triples satisfying 2^pU−3H=1. Different members share this defect value while retaining different sources.

Main use: Section 2 · Contents

Constitution and Mountain

K_M specifies the membership, address and member constructors, boundary, identity, common fiber and provenance of the whole family. Mountain is that constituted Macro object. See the Macro tensor entry and E10.

Main use: Section 4 · Contents

Fold and Unfold

Fold_M carries Mountain to the typed pair $\langle 1, D_C^{-1}(\{1\})\rangle$. Unfold_M returns the constituted family only if lawful admission fixes its enrichment or the reference uniquely determines K_M, addresses and provenance in the admitted recovery environment. The extensional arithmetic fiber alone is insufficient; matching the inherited recovery contract remains open in Appendix B. The scalar 1 alone is insufficient. See E11.

Main use: Section 4 · Contents

Emergent One

The returned presentation E_H retaining Templex One, Q_H and its witness W_H. Its typed agreement with Templex One and its scalar fold value are separate statements. See E15 and E16.

Main use: Section 6 · Contents

VPA

The execution sequence $V\to P\to A\to V'$ and the cyclic role action $\rho(V)=P$, $\rho(P)=A$, $\rho(A)=V$. It acts within the scale construction. Squadron presents the scales together and supplies no additional executor. See E13.

Main use: Section 5 · Contents

ENAID

The recursively addressed population and cross-scale extension of the VPA constitution used by the inherited projection and address rules. In this paper its all-number projection remains an explicit premise. See E09 and Appendix B.

Main use: Section 4 · Contents

Defect

$D_C(H,p,U)=2^{pU}-3H$. On a reconstructed odd-source triple it equals 1. This is an algebraic residual, distinct from an orbit stopping time. See E04.

Main use: Section 2 · Contents

AMM witness

W_H retains the certified Atomic–Micro–Macro return data, the packet, its defect certificate, address and provenance. Its full inherited return premises are distinguished from the five checked packet declarations. See E15.

Main use: Section 6 · Contents

Appendix H. Internal evidence addresses

Public links below identify repository records at the stated edition or commit. Records labeled internal are retained in the author's research archive; they have no public retrieval address established in this review and are not delivered with this paper. Their identifiers locate the editorial record, not an independently accessible proof. The cited sections reproduce the relevant summary and its limitations. Original author-source bodies I1–I5 likewise remain archival sources; their recovery limits are stated in Appendix B.

P1. Historical edition citation metadata. CITATION.cff, version publication-2026-09-10. This identifies the earlier research package, not this paper.

P2. GitHub release metadata. Release v1.0.0, release ID 386819774, published 2026-09-11T06:12:04Z.

P3. Retrieved tag reference. Tag v1.0.0 resolves in the saved record to commit 0e5f0acd. This is distinct from the later submission commit.

P4. Retrieved commit metadata. Commit c80c8245, "Refresh research storefront cross-links," recorded 2026-09-13T22:11:05Z.

P5. Authorship and contribution register. Internal editorial record COLLATZ_GATE_3_1 / AUTHORSHIP_AND_RIGHTS_REGISTER.md; author and assistance summary appears in "Authorship, assistance and provenance."

P6. Source and publication chronology. Internal editorial record COLLATZ_GATE_3_2 / CHRONOLOGY.md, review dated October 6, 2026; the paper's provenance section summarizes its event distinctions.

V1. Exact Lean coverage and saved checker records. Internal review COLLATZ_GATE_4_2 / LEAN_COVERAGE.md. Public formal source and historical verification record identify the checked project. Section 8 and Appendix C delimit the five Snowman targets and historical run.

V2. Bounded replay and execution receipt. Public replay and result. Internal rerun receipt COLLATZ_LOCKED_PERFORMANCE_261006 / PERFORMANCE_RECEIPT.json records 2026-10-07T01:16:52Z, exit zero, unchanged published replay.

V3. Beta 3 finite model receipt. Internal development record COLLATZ_HOMEWORK_BETA_3 / REPLAY_RECEIPT.json; section 8 and Appendix D state the populations and model limits.

V4. Dependency and circularity map. Internal editorial record COLLATZ_GATE_2_4 / DEPENDENCY_AND_CIRCULARITY_MAP.md; Appendix B states the inherited interfaces and unresolved source matching.

V5. Replay/control review and proposed formalization. Internal editorial record COLLATZ_GATE_4_7 / REPLAY_AND_FORMALIZATION_REVIEW.md; Appendix C separates proposed targets from checked declarations.

V6. Finite/symbolic reconciliation and historical failure locators. Internal editorial record COLLATZ_GATE_4_5 / EVIDENCE_RECONCILIATION.md; Appendix D records the reported failures and unavailable original packets.

V7. Cross-domain bridge review and source pins. Internal editorial record COLLATZ_GATE_4_8 / BRIDGE_REVIEW.md; Appendix E states the topology convention and its authority limit.

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Author’s source manuscripts and formalization

I1. Gattenby, Virgil Lee / TRYX-ENAID program (24 August 2026). *Existence of Templex One: algebraic, Micro, Macro and rotation closure*. Canonical theorem manuscript, identifier ending 260824.1407.

I2. Gattenby, Virgil Lee / TRYX-ENAID program (24 August 2026). *Canonical algebraic, Micro, Macro and Squadron lock manifest*. Identifier ending 260824.1409.

I3. Gattenby, Virgil Lee / TRYX-ENAID program (24 August 2026). *Collatz internal solution statement*. Identifier ending 260824.1434.

I4. Gattenby, Virgil Lee / TRYX-ENAID program (1 September 2026). *Snowman continuity closure: reintroduction benchmark*. Identifier ending 260901.003644Z.

I5. Gattenby, Virgil Lee / TRYX-ENAID program (5 September 2026). *Historical internal closure benchmark lock*. Identifier ending 260905.

I6. TRYX project. *TryxProof.lean*, namespace TRYX.Snowman. Formal source at commit c80c82450cbbeb9148f5a78189b307a3336111fd; lines 87–144.

The author-source titles above are descriptive short titles; their dated identifiers distinguish the archived bodies. Appendix H identifies the P/V documentary records and their access status. Internal source and review files are not distributed with this standalone paper.